Trey Smith

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GitHub: https://github.com/Treys925

EDUCATION

The University of Michigan — Ann Arbor, MI
B.S., Philosophy (Honors) — April 2021
B.S., Mathematics — April 2021

RESEARCH EXPERIENCE

The Institute for Social Research — Ann Arbor, MI
Bioinformatician, — Summer 2023– Present
Survey Research Center PNG

- DNA Methylation (DNAm) QC and analyses for the Health and Retirement Study (HRS), Future of Families and Child Wellbeing Study (FFCW), Indiana Person to Person Study (P2P), and Michigan Twin Study (MTwiNS)
- Analysis and modeling of DNAm, Genetic, RNA-seq, and Neuroimaging data in R
- Contributed to several publications, including as first author
- Trained PhD students and Postdocs to use R scripts for DNAm analysis pipelines in the GeSSe course

The Institute for Social Research — Ann Arbor, MI
Research Assistant II, — Summer 2022 - Summer 2023
ISR BioSpecimen Lab

- Project lead for HRS Venous Blood Study (VBS) and the Study of Adolescent Neurodevelopment (SAND) collection kits, performed dried blood spot card assessment and punching
- Ran Epigenome Wide Association Studies (EWAS) on HRS and FFCW DNAm data
- Contributed to laboratory protocols for sample processing and freezer storage

University of Michigan — Ann Arbor, MI
Student Researcher, — Winter 2020 – Summer 2021
Michigan Lab of Geometry

- Implemented and tested 3 versions of Conway's game of life in Python and Matlab
- Developed unique rules using Fourier transformations and stochastic models
- Presented at the Lab of Geometry poster session

The Institute for Social Research — Ann Arbor, MI
Research Assistant, — Summer 2019 – Fall 2020
Survey Research Center

- Performed data analysis and review for HRS and Consumption and Activities Mail Survey (CAMS) using SAS software

- Coded survey responses for the 2018 HRS study
- Data entry for CAMS project

University of Michigan Ann Arbor, MI
Research Assistant, Fall 2018 – Summer 2019
Bigham Lab for Anthropological Genomics

- Ran Polymerase chain reactions, Gel-electrophoresis for sequencing
- Developed statistical analyses of data to test proposed biological feature
- Presented at the American Association of Physical Anthropologists

AWARDS AND HONORS

University of Michigan Honors	2021
University of Michigan Honors	2019
Michigan Anthropology Undergraduate Award, Department of Anthropology Curriculum Committee -- Daniel Carl Maier Fund	2019
University of Michigan Honors	2017

GRANT SUPPORT (AS BIOINFORMATICIAN)

NIH/NIA "Health and Retirement Study: Yrs 35-40" 05/2024 - 12/2029
(Weir, PI)
U01 AG009740-35
Total Award Amount (including Indirect Costs): $195,122, 882

NIH/NIA "Genomics for Social Scientists: 2022-2027" 08/2022 - 04/2027
(Faul/Mitchell/Ware, MPI)
R25 AG053227-08
Total Award Amount (including Indirect Costs): $3,009,650

NIH/NIA "Network on Biomarker Collection and Measurement in Population Studies of Aging" 06/2022 – 03/2027
(Faul/Crimmins/Mitchell, MPI)
R24 AG054365-07
Total Award Amount (including Indirect Costs): $1,972,145

NIH/NIA "Biological Aging Across the Life Course: Harmonizing Cohort Biospecimen Archives" 03/2021 – 02/2027
(Faul/Mitchell, MPI)
R01 AG071071-02
Total Award Amount (including Indirect Costs): $599,164

NIH/NIA "Social Circumstances and Epigenomics Promoting Health 09/2020 - 04/2026

in Three Countries"
(Faul/Crimmins, MPI)
R01 AG068937-01
Total Award Amount (including Indirect Costs): $259,664

NIH/NIA "Biological Underpinnings of Socioeconomic Differentials in Health and Mortality" 09/2018 - 06/2024
(Crimmins, PI)
R01 AG060110-02
Total Award Amount (including Indirect Costs): $716,321

PEER-REVIEWED PUBLICATIONS

1. Ryan Tung, Carly O'Neill, Trey Smith, Felicia A. Hardi, Leigh G. Goetschius, Edward D. Huntley, Helen C.S. Meier, Luke W. Hyde, Dan Notterman, Christopher S. Monk, Colter Mitchell, Associations between surrogates of prenatal lead DNA methylation and adolescent neural network architecture, *Brain and Environment*, Volume 5, 2026, 100013, ISSN 3050-5917, https://doi.org/10.1016/j.braen.2026.100013

2. deSteiguer, A.J., **Smith, T**., Goode, J.A. *et al.* Salivary DNA methylation and pubertal development in adolescents. *Sci Rep* **15**, 35970 (2025). https://doi.org/10.1038/s41598-025-19588-1

3. Lussier, A. A., Smith, B. J., Fisher, J., Luo, M., Cerutti, J., Schneper, L., **Smith, T**., Cecil, C. A. M., Felix, J. F., Mitchell, C., Notterman, D. A., Ressler, K. J., Schaid, D. J., Simpkin, A. J., Suderman, M. J., Walton, E., Smith, A. D. A. C., & Dunn, E. C. (2024). DNA methylation mediates the link between adversity and depressive symptoms. *Nature Mental Health*, 2(12), 1476-1485. Article 134. https://doi.org/10.1038/s44220-024-00345-8

MANUSCRIPTS UNDER REVIEW

1. Wu, Q. , de Steiguer, A. , Zarandooz, S., Willems, Y.E. , **Smith, T**. , Goode, J. ,Mitchell, C., Harden, K. P., Raffington, L. (2025). Epigenetic Scores of Accelerated Pubertal Development are Associated with Aging-Related Health Outcomes among Older Adults. Under Review

2. Jessica Faul, Stacey Collins, **Trey Smith**, Eric T. Klopack, Colter Mitchell, Eileen M Crimmins, and Mateo P Farina. (2025). Applying epigenetic g as a biomarker of cognitive aging in a diverse, nationally representative sample of older adults. Under Review

MANUSCRIPTS IN PROGRESS

1. **Trey Smith**. (2026). CoMeBackV2: An R package for Co-Methylated regions on Illumina arrays.

2. Lisha Lin, Helen Meier, **Trey Smith**, Joshua Goode, Jessica Faul, Daniel A. Notterman, Jennifer A. Smith, Colter Mitchell. (2025). Time-Sensitive Associations Between Early Life Poverty and the Epigenome in Children and Adolescents from the Future of Families and Child Wellbeing Study.

3. **Trey Smith**, Thomas Karadimas, Farah Ammous, Helen Meier, Colter Mitchell, Jessica Faul. (2025). BMI poly-epigenetic scores predict incident comorbidities and mortality in a population representative sample.

OTHER PUBLICATIONS

1. Seshadri, G., **Smith, T**., Klopack, E., Crimmins, E., Thyagarajan, B., Faul, J. HRS Documentation Report, *RNA SEQ (Count Data) 2016 Health and Retirement Study Venous Blood Study (VBS)*, Dec 2025.
2. **Smith, T**., Crimmins, E., Faul, J. HRS Documentation Report, *HRS Epigenetic Clocks – Supplement PACE and Grim2*, Dec 2025.
3. **Smith, T**. , Ozer, Aksel. (2025). Linear Orders and the Real Line. arXiv preprint arXiv: 2508.15644
4. **Smith T** (2025). _CoMeBackV2: CoMeBackV2: Co-Methylation With Genomic CpG Background for EPICv2. R package version 0.1.0, commit 314b8a07cb802f7654dcca090c4943f9417dce99, <https://github.com/Treys925/CoMeBackV2_R_package>.
5. Desteiguer, A. J., **Smith, T**., Goode, J., Willems, Y., Tucker-Drob, E. M., Raffington, L., ... & Harden, K. P. (2024, November). Epigenome-Wide Analyses of Pubertal Development in Adolescence: Connections with Biological Aging. In BEHAVIOR GENETICS (Vol. 54, No. 6, pp. 519-519). ONE NEW YORK PLAZA, SUITE 4600, NEW YORK, NY, UNITED STATES: SPRINGER.
6. **Smith T**, Karadimas T, Meier H, Mitchell C, Faul J. BMI POLYEPIGENETIC SCORES PREDICT FUTURE CHRONIC CONDITIONS AND CHOLESTEROL. Innov Aging. 2023 Dec 21;7(Suppl 1):219. doi: 10.1093/geroni/igad104.0723. PMCID: PMC10737489.
7. **Smith, T**., (2021), *Evolutionary Epistemology: Popper, Peirce, and Multi-level Selection*, [Bachelor's Thesis, University of Michigan]

PRESENTATIONS

Smith, T., Mitchell, C., Peterson, M., Faul, J. Understanding the Speed of Aging Among Adults with Cerebral Palsy. 2026 UMAging Initiative Symposium, *Stress, Resiliency, and Health in Later Life*, speaker, Ann Arbor, MI, March 2026.

deSteiguer, A.J. Wu, Q., **Smith, T**., Goode, J., Tucker-Drob, E.M., Mitchell, C., Harden, K.P., Raffington, L. (2025, November 13). DNA-methylation Patterns of Socioeconomic Status and Poverty in Children's Saliva. Talk presented at the 58th Annual Meeting of the International Society for Developmental Psychobiology, San Diego, CA.

Sullivan SM, Silkin C, **Smith T**, Guan W, Faul JD, Thyagarajan B. Sex-specific differences in epigenetic-based biomarkers of lead exposure and kidney function among a representative sample of older Americans, International Society for Environmental Epidemiology, Aug 2025.

Goode J, **Smith T**. MethylSurroGetR: Tools to Generate Predicted Values of DNAm Surrogates in R, 2025 Genomics for Social Scientists Epigenetics Workshop, June 2025.

Ammous F, **Smith T**, Mitchell C. Adolescent body mass index and its association with DNA methylation: Findings from the Future of Families and Wellbeing Study, Society for Epidemiologic Research Annual Meeting, June 2025.

Goode J , **Smith T**, Faul J, Notterman D, Mitchell C. DNA Methylation as a Surrogate Biomarker for Early-Life Poverty, Population Association of America 2025 Annual Meeting. Washington D.C. April 2025.

Sullivan SM, Silkin C, **Smith T**, Guan W, Faul JD, Thyagarajan B. Sex-specific differences in epigenetic-based biomarkers of lead exposure and kidney function among a representative sample of older Americans, Molecular Pathology and Genomics Symposium, Feb 2025.

Smith, T. Gene Enrichment, 2025 Genomics for Social Scientists Epigenetics Workshop, speaker, Jan 2025.

Desteiguer, A. J., **Smith, T**., Goode, J., Willems, Y., Tucker-Drob, E. M., Raffington, L., ... & Harden, K. P. , Epigenome-Wide Analyses of Pubertal Development in Adolescence: Connections with Biological Aging, Behavioral Genetics Nov 2024.

Mitchell C, **Smith T**, Goode J, Ware E, Meier H, Faul J, Notterman D. Utilizing Longitudinal Methylation Analyses to Connect Environment to Health, Development, and Aging, Population Association of America 2024 Annual Meeting. Columbus, OH. April 2024.

Smith T, Karadimas T, Meier H, Mitchel CM, Faul J. BMI Polyepigenetic Scores Predict Future Chronic Conditions and Cholesterol, Gerontological Society of America 2023 Annual Meeting, speaker, Tampa, FL. November 2023.

Smith, T., "Evolutionary Epistemology: Popper, Peirce, and Multi-level Selection", LSA Philosophy Honors Thesis Defense, speaker, Ann Arbor, MI, 2021

Du, E., Jackson, F., Ke, D., **Smith, T**., Riley, B., Sprunger, C., Storey, K., Zimmerling, J., "Extensions to Conway's Game of Life", Michigan Lab of Geometry Poster Conference, speaker, Ann Arbor, MI, 2021

Smith, T., Wang, K., "High-Altitude adaptation in gene H19," American Association of Physical Anthropologists 88th conference, speaker, Cleveland, OH, 2019

TEACHING EXPERIENCE

Genomics for Social Scientists – Epigenetics (GeSS-Epi) – University of Michigan (Instructor), Jan 2026

Genomics for Social Scientists – Epigenetics (GeSS-Epi) – University of Michigan (Instructor), June 2025

Genomics for Social Scientists – Epigenetics (GeSS-Epi) – University of Michigan (Instructor), Jan 2025

Genomics for Social Scientists (GeSS) – University of Michigan (Instructor), June 2024

PROFESSIONAL SERVICES

Reviewer for Journal of Open Source Software (JOSS), 2024 – Present

Abstract Reviewer for Gerontological Society of America Conference, 2024

Editor for the Meteorite Undergraduate Philosophy Journal, 2021

REFERENCES

Colter Mitchell, ISR
Research Associate Professor, Survey Research Center, Supervisor
cmsm@umich.edu

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Research Associate Professor, Survey Research Center, Supervisor
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